

Read quality and genome alignment in the PDAC KRAS dataset

WEEK 2 PROGRESS REPORT

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Illustrated version: full alignments, quality checks and calculation tables.

Wording revised on 23 September 2026. Results retain the original reporting stage.

Methods

FastQC 0.12.1 assessed the 34 input files. MultiQC 1.35 combined 32 Galaxy reports with two reports from the complete local SRR24828471 files. HISAT2 2.2.3 aligned the paired reads to human hg38 with Phred+33 quality scores, unstranded mode, no read limit and no end trimming. Other alignment options used the defaults.

Incomplete imports were replaced before analysis. Base-quality and adapter checks passed, so Cutadapt was not used. The paired files remained grouped by sample and run accession.

Quality and strand results

All 34 files contain 150-base reads and pass base-quality and adapter checks. Base composition and duplication are flagged in every file. MultiQC reports 57.0%-67.3% duplication. Both SRR24828471 mates also have GC-distribution warnings.

Library preparation can affect base composition, while abundant transcripts and amplification can produce repeated sequences. The quality flags do not identify their cause. No trimming or automatic duplicate removal was applied on the basis of these flags. The GC warning alone does not establish contamination.

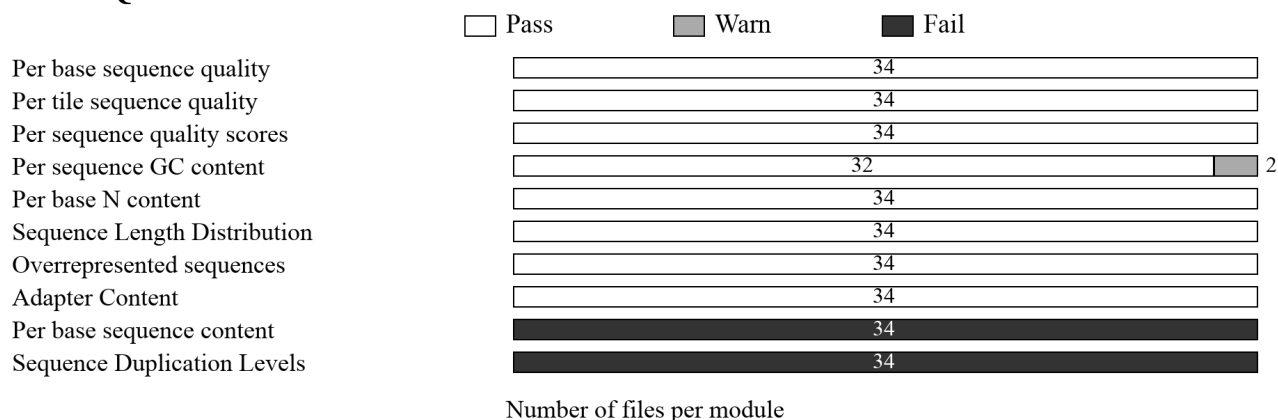
Pilots used the first million pairs from Control M1 and KRAS P1. Overall alignment was 96.63% and 96.40%, respectively, under unstranded, forward-stranded (FR) and reverse-stranded (RF) settings. Unique-pair alignment was 86.40% versus 86.44% (FR/RF) for Control M1 and 87.43% throughout for KRAS P1.

RSeQC Infer Experiment sampled 200,000 reads with a minimum mapping quality of 30 using an hg38 BED12 annotation. The two orientation fractions were 46.90% and 46.73% for Control M1, and 45.80% and 45.69% for KRAS P1. These similar fractions support unstranded alignment for the two inputs. The remaining runs used this setting without individual strand checks.

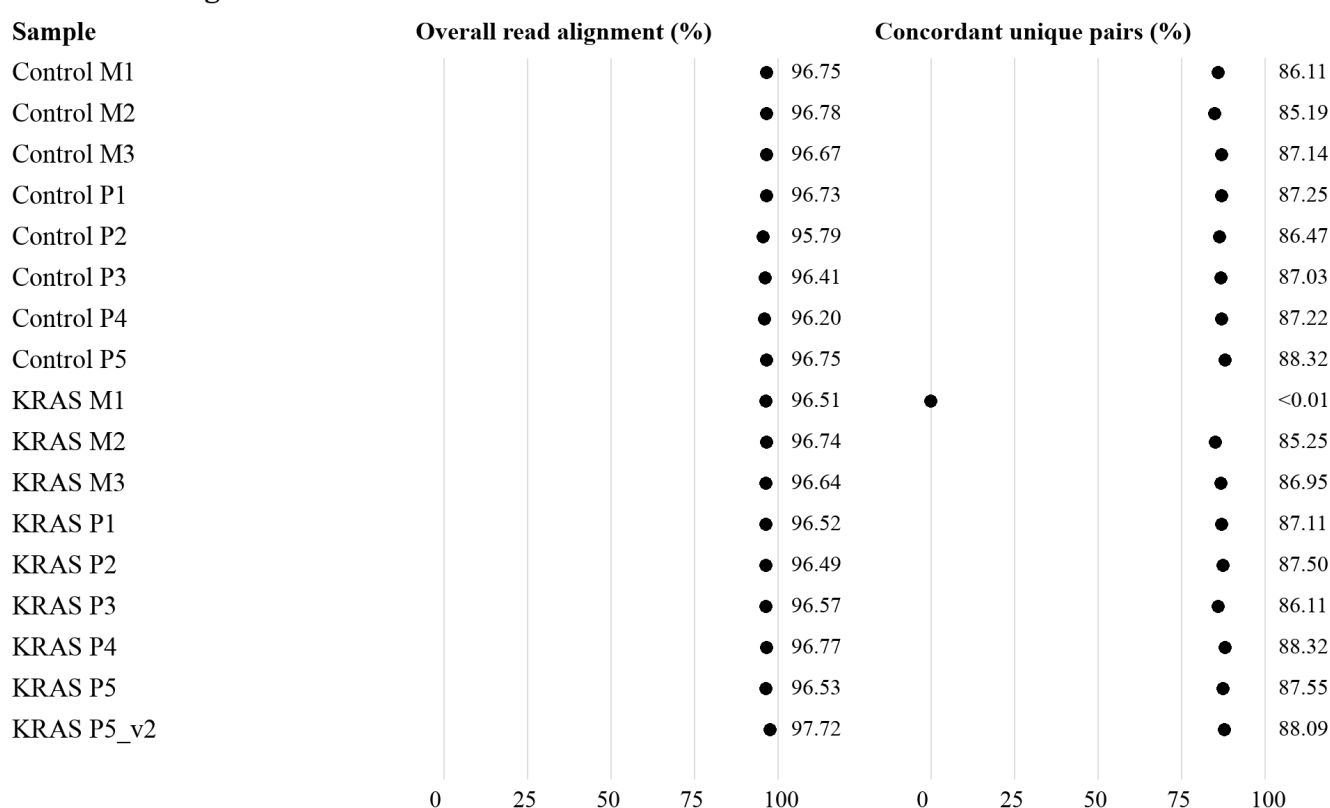
Full-run overall alignment ranges from 95.79% to 97.72%. KRAS M1 has only 0.0030% concordant unique pairs and 75.56% discordant pairs, so it needs review before counting genes. Gene counts and differential expression are not yet available.

Read quality and alignment figures

A FastQC module results across 34 files



B Full run alignment rates across 17 runs



Source: verified FastQC reports and full HISAT2 summaries, 22 September 2026.

Figure 1. A: FastQC results for 34 input files. B: overall alignment and concordant unique-pair alignment for all 17 full runs. Overall alignment uses individual reads; unique-pair alignment uses input pairs. The two measures differ most strongly for KRAS M1.

Appendix A. Full alignment results

Sample	Input pairs	Overall (%)	Unique pairs (%)	Discordant (%)
Control M1	40,237,116	96.75	86.11	0.61
Control M2	45,764,716	96.78	85.19	0.58
Control M3	38,181,985	96.67	87.14	0.64
Control P1	45,946,458	96.73	87.25	0.57
Control P2	39,989,452	95.79	86.47	1.15
Control P3	47,840,900	96.41	87.03	0.54
Control P4	46,287,075	96.20	87.22	0.65
Control P5	45,023,981	96.75	88.32	0.57
KRAS M1	45,292,678	96.51	0.0030	75.56
KRAS M2	44,032,049	96.74	85.25	0.62
KRAS M3	42,961,283	96.64	86.95	0.60
KRAS P1	39,963,282	96.52	87.11	0.62
KRAS P2	42,632,214	96.49	87.50	0.69
KRAS P3	42,073,244	96.57	86.11	0.56
KRAS P4	40,452,536	96.77	88.32	0.52
KRAS P5	37,886,040	96.53	87.55	0.64
KRAS P5_v2	43,166,706	97.72	88.09	0.32

Overall (%) = $100 \times [1 - \text{unmapped reads} / (2 \times \text{input pairs})]$.

Unique (%) = $100 \times \text{concordant unique pairs} / \text{input pairs}$.

Discordant (%) = $100 \times \text{discordant unique pairs} / \text{input pairs}$.

For Control M1: overall alignment = $100 \times [1 - 2,617,722 / (2 \times 40,237,116)] = 96.75\%$. Unique-pair alignment = $100 \times 34,649,263 / 40,237,116 = 86.11\%$.

Total input: 727,731,715 pairs. KRAS P5 is SRR24828471; P5_v2 is SRR24828472. The two runs remain separate while their repeat type is unresolved.

Appendix B. FastQC results

FastQC check	Pass	Warn	Fail	Files
Per-base quality	34	0	0	34
Per-tile quality	34	0	0	34
Per-read quality	34	0	0	34
Base composition	0	0	34	34
GC distribution	32	2	0	34
N content	34	0	0	34
Read length	34	0	0	34
Duplication	0	0	34	34
Overrepresented sequences	34	0	0	34
Adapters	34	0	0	34

Each row covers 34 files. For GC distribution, 32/34 passed (94.12%) and 2/34 received warnings (5.88%). A module flag describes a quality measure; it does not mean the FastQC job failed.

References

1. Galaxy FastQC reports, MultiQC 165, all 17 full HISAT2 summaries and RSeQC results, recorded on 22 September 2026. Analysis records: github.com/yichuanlin07/pdac-kras-rnaseq.
2. Course slides: Week 2, Quality Control and Alignment.
3. Klomp JA et al. Defining the KRAS- and ERK-dependent transcriptome in KRAS-mutant cancers. Science. 2024. doi:10.1126/science.adk0775.
4. FastQC documentation. bioinformatics.babraham.ac.uk/projects/fastqc/.
5. HISAT2 manual. daehwankimlab.github.io/hisat2/manual/.
6. RSeQC documentation. rseqc.sourceforge.net.

Author contribution

Yichuan Lin organized the data and coordinated the analysis and report preparation.

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